PSB605IT – Individual Computing Science Project

Drive Buddy - A Machine Learning – Enhanced Drowsiness Detection and Monitoring System

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Problem Statement

According to Singapore Medical Journal (Lim & Chia, 2015), driver fatigue is one the biggest health and safety concern with the road transport sector after surveying 340 taxi drivers.

1.5K	Fatalities involving drowsy driving annually.
71K	Injuries caused annually cause by drowsy driving.
$12.5B	Economic loss attributed to drowsy driving accidents.
1	One life cost to many.

This numbers demands effective intervention to mitigate the far-reaching economic, social, and human consequences associated with drowsy driving.

Dozens of research in areas of image recognition using state of the art pattern recognition are available online but very few to none cost-effective solutions are available to life-line consumers.

Solution Brief

Drive Buddy, is a co-pilot safety system employing Eye Aspect Ratio (EAR) Logic and Machine Learning to detect drowsiness through EAR analysis. Using Machine Learning and Image Recognition system, the solution aims detect fatigue markers as well other eye distractions.

Features

- Measure Eye Aspect Ratio using ML model to identify key eye markers
- Identify whether EAR is above threshold and trigger preset interventions
- Provide Audio and Visual monitors if drowsiness pattern is detected
- Initiate External Party Intervention via Email and SMS if on-board interventions are ignored and persistent threshold breach is detected
- Online analytics to monitor and review activities

Use Cases

- Road Driving
- Machinery Operations (Trains, Cranes, Heavy equipment, etc)

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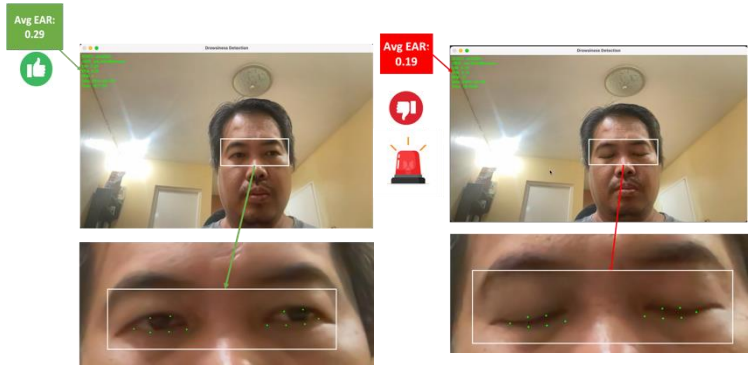
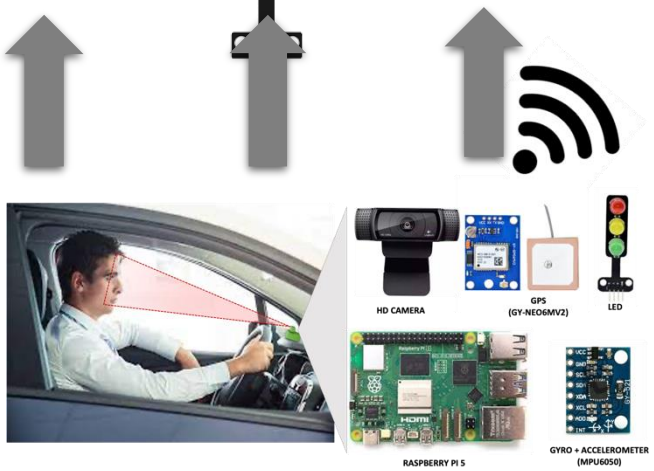
Drive Buddy Operation

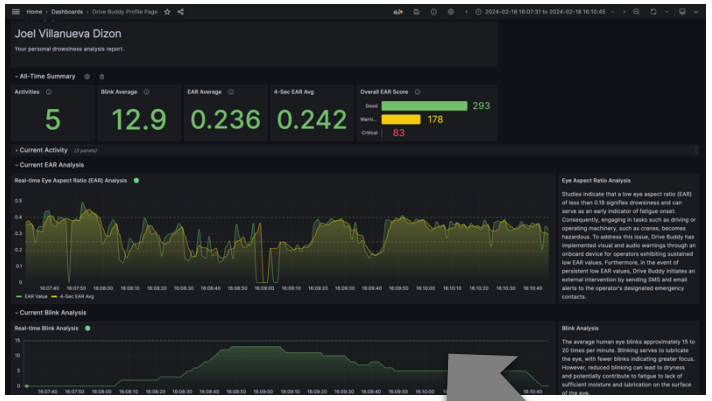


On-Board Interventions

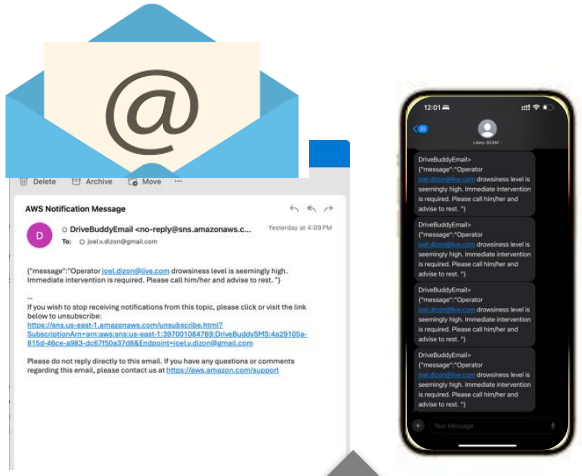
Continuous Monitors (Visual)

Audio Intervention (4-Sec Avg above threshold)



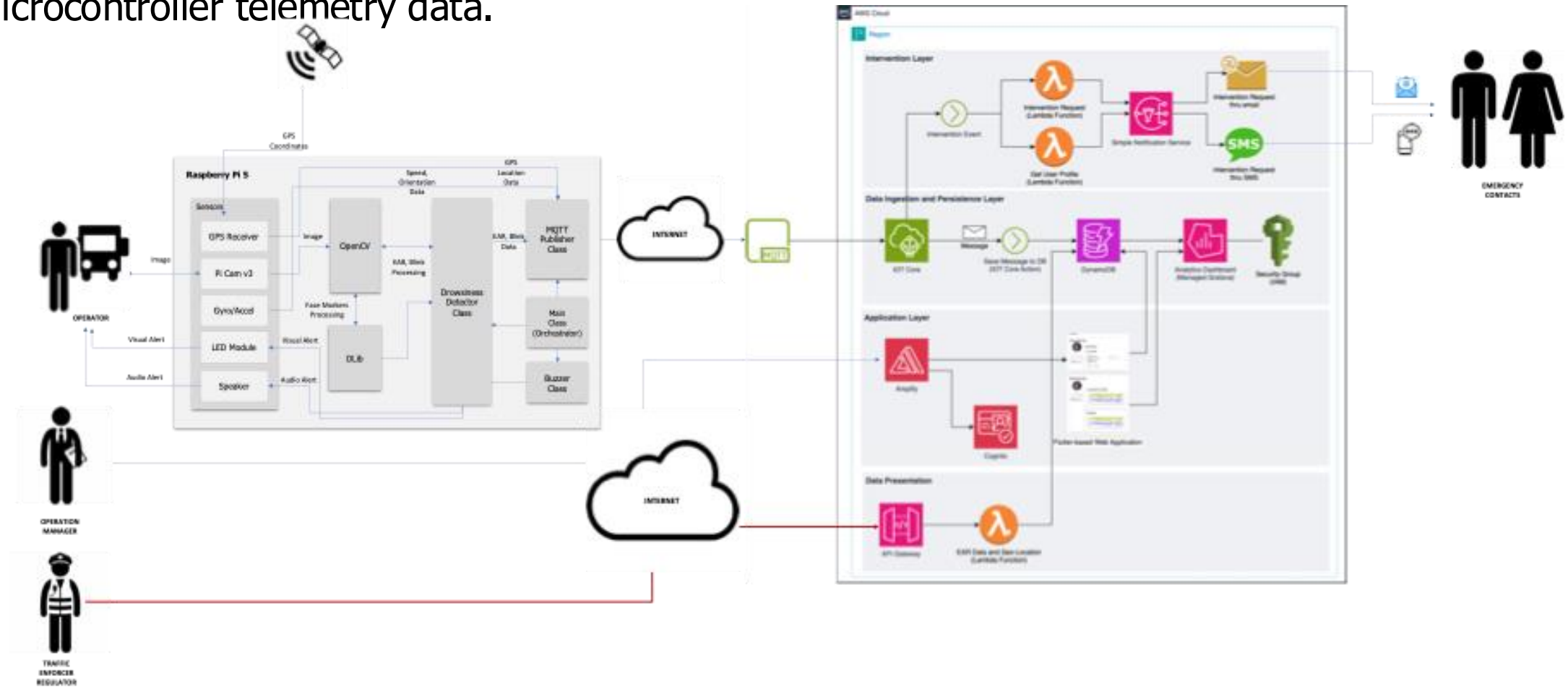


External Intervention Request via SMS/Email (4-Sec Avg above threshold)

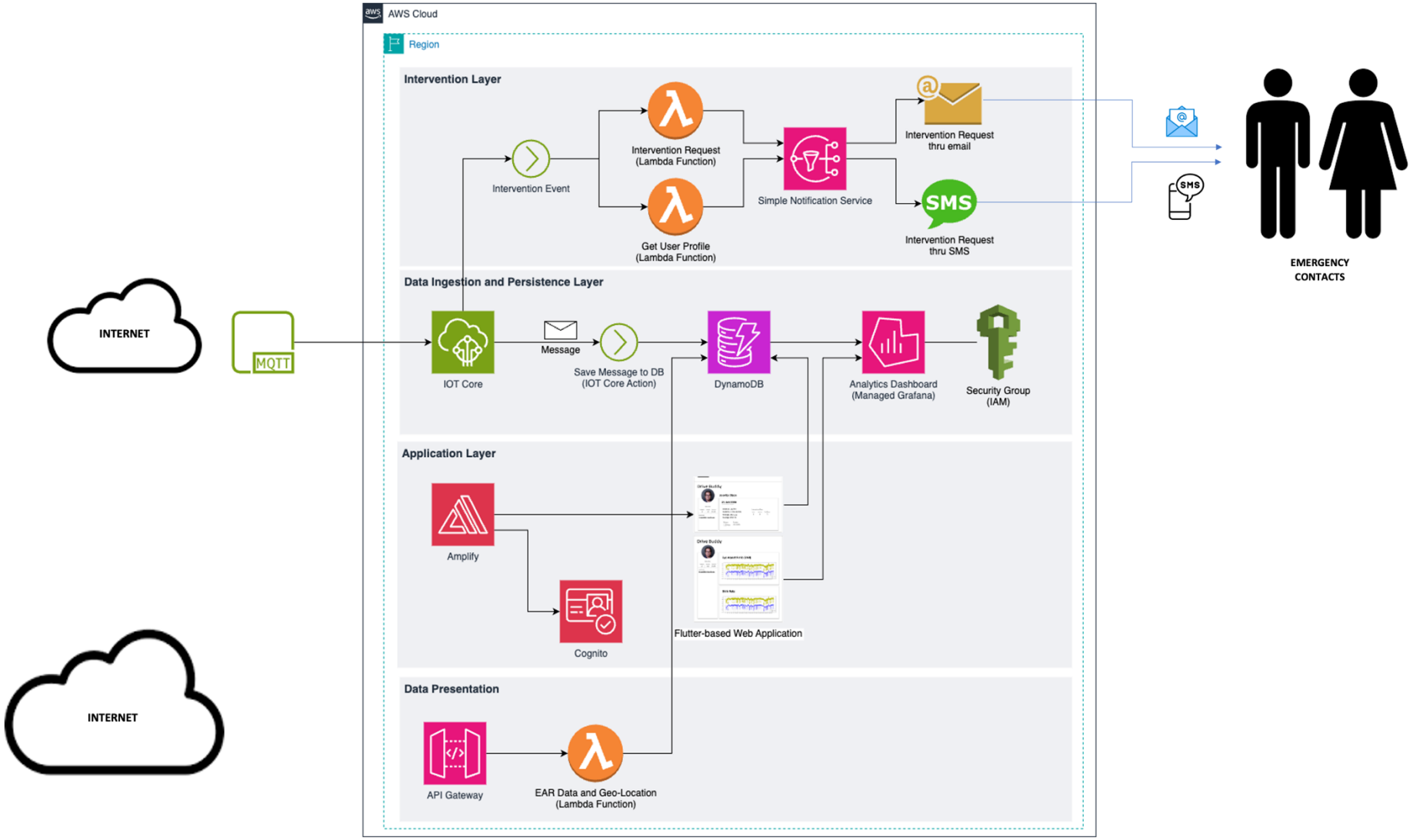


Solution Architecture

The Drive Buddy solution comprises two main components: the frontend and backend tiers, leveraging various AWS services to manage, process, and secure telemetry data. The frontend acquires data such as eye tracking, operator identity recognition, speed, geolocation, and other Microcontroller telemetry data.



Prototype Design (Cloud-Side)



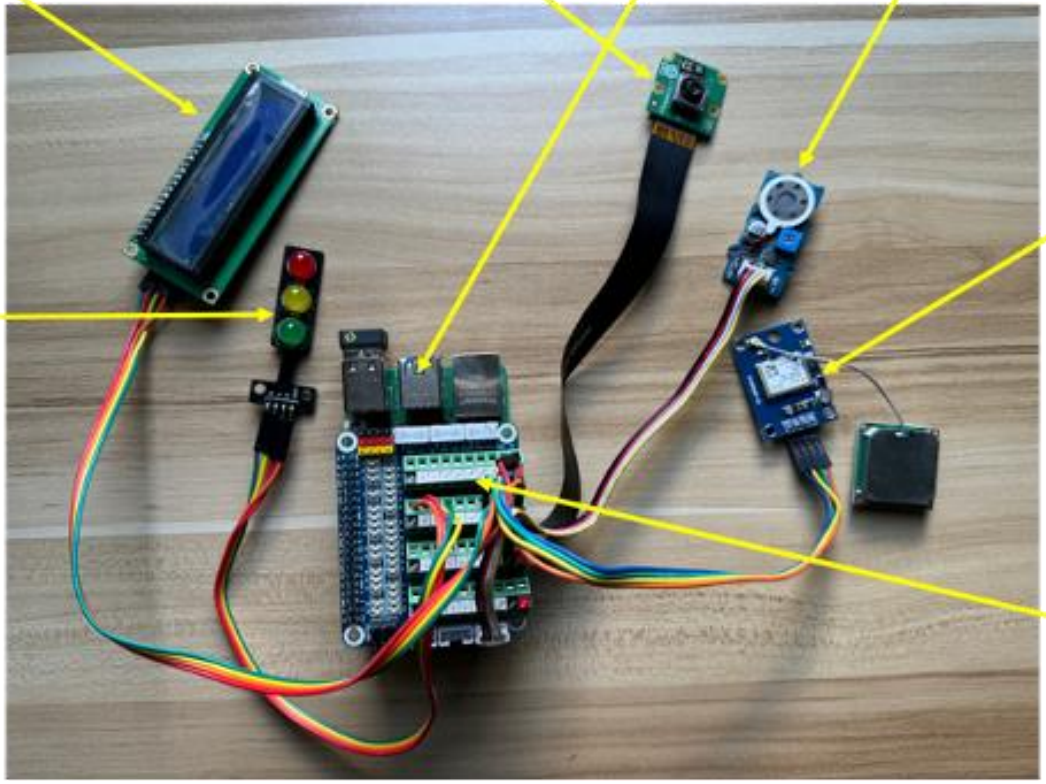
Prototype Design (Device-side)

Raspberry Pi Cam v3
Latest Sony camera sensor with 75 degree narrow field of view. F/1.8 aperture for low light image capture.

Raspberry Pi 5
Microprocessor to control all sensors and business logic of Drive Buddy system. Use Debian 12 Bookworm OS.

Speaker Module
For Audio intervention if Avg EAR goes below 0.20.

I2C 16x2 LCD
Provide visual measures of EAR, Avg EAR, Average Blink



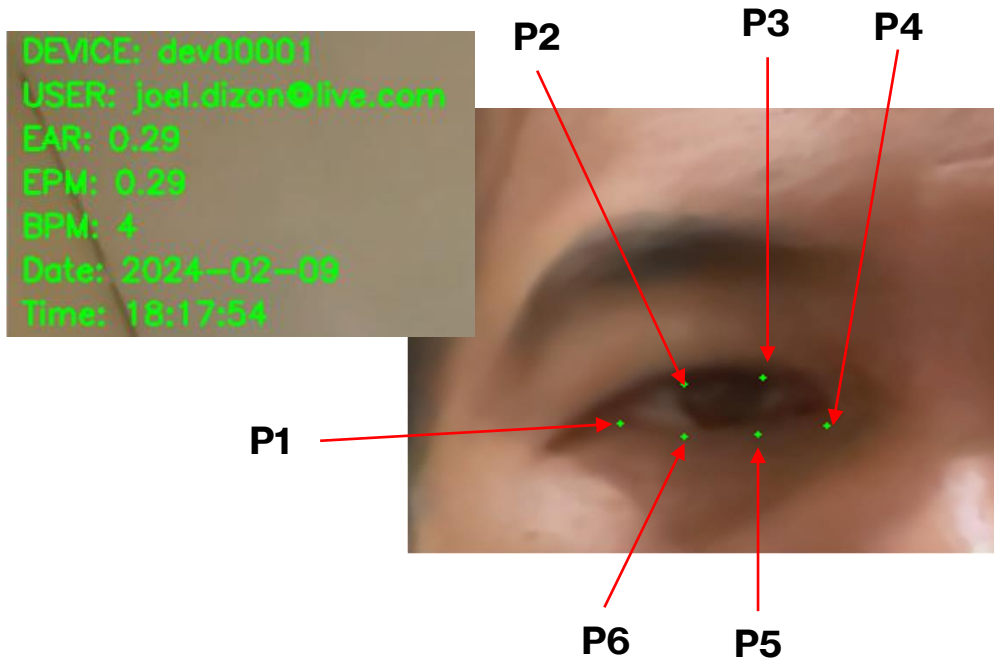
GPS Module
GPS Receiver for location tracking.

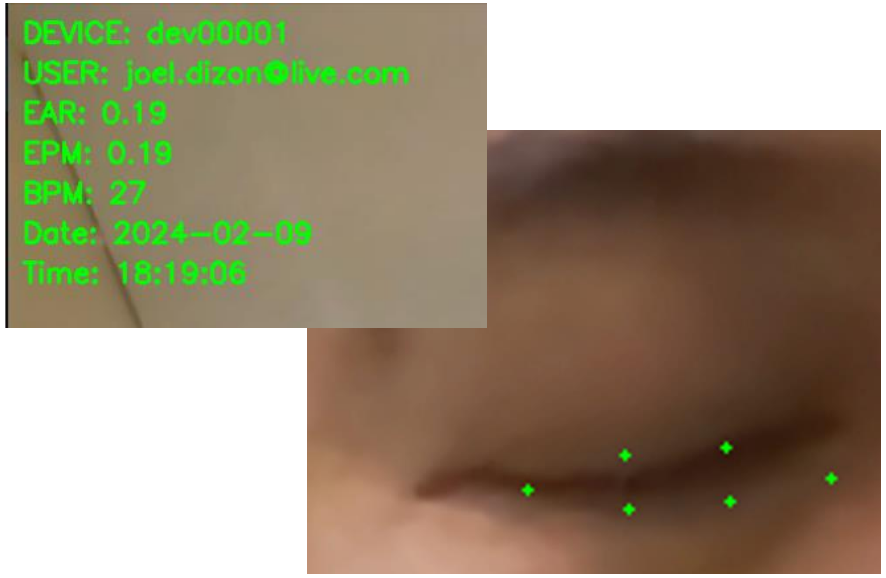
Traffic Light Module
Provide visual alert level of Avg EAR (Green = Good, Yellow = Warning, Red = Critical)

GPIO Expansion HAT
For easy sensor integration. Offers stable and robust sensor connectivity .

Eye Aspect Ratio (EAR)

Measurement used to determine the extent to which the eyes are open or shut.





P1 – P6 represent the coordinates (in pixels) of the six eye landmarks

*P*1=(*x*1,*y*1) for the left corner of the eye.
*P*2=(*x*2,*y*2) for the top-left point of the eye.
*P*3=(*x*3,*y*3) for the top-right point of the eye.
*P*4=(*x*4,*y*4) for the right corner of the eye.
*P*5=(*x*5,*y*5) for the bottom-left point of the eye.
*P*6=(*x*6,*y*6) for the bottom-right point of the eye.

$$\text{EAR} = \frac{\text{Distance between vertical landmarks}}{\text{Distance between horizontal landmarks}}$$

$$\text{EAR} = \frac{(P2 - P6) + (P3 - P5)}{2\,(P1 - P4)}$$

Prototype Demo

DEV Environment
Mac

DEV Environment
Raspberry Pi

Visualizations
https://g-ef00e70ebd.grafana-workspace.us-east-1.amazonaws.com/d/P_ueo32Sk/drive-buddy-profile-page?orgId=1&from=now-5m&to=now&refresh=5s

Conclusion

The delivery of the Drive Buddy final project marks a significant milestone for me, symbolizing the culmination of my education at PSB and Coventry University.

While currently a prototype, it shows immense potential for refinement and enhancement. Through this project, I've demonstrated how to leverage on existing research and technology to develop a high-quality, low-cost solution that may rival leading brands, and potentially accessible to lifeline users.

This innovation heralds a new era in safety, potentially saving countless lives and millions of dollars.

Moving forward, ongoing improvements are imperative, including enhancing user experience, expanding features, ensuring data security, optimizing performance, and addressing physical nuances. These discussions and refinements are pivotal in maximizing the project's impact and realizing its full potential.